

## **Effects of consonant-conditioned informativity on vowel duration in Japanese**

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### **Abstract**

Research on English and other languages has shown that syllables and words that contain more information tend to be produced with longer duration (e.g. Aylett & Turk 2004). This research is evolving into a general thesis that speakers articulate linguistic units with more information more robustly. While this hypothesis seems plausible from the perspective of communicative efficiency, previous support for it has come mainly from English and some other Indo-European languages. Moreover, most previous studies tended to focus on rather global effects, such as the interaction of word duration and sentential/semantic predictability, but we feel that it is also essential to explore a more local interaction between adjacent segments, where meaning is irrelevant. With these two issues in mind, the current study examines the effects of local informativity on vowel duration in Japanese, using the Corpus of Spontaneous Japanese (the CSJ). To examine consonant-vowel phonotactics within a CV-mora, consonant-conditioned Shannon entropies were calculated, and their effects on vowel duration were examined, together with other linguistic factors that are known from previous research to affect vowel duration. In addition to confirming several linguistic factors affecting vowel duration, the current study reveals rather complex effects of consonant-conditioned entropy on vowel duration in Japanese.

**Keywords:** informativity, entropy, vowel duration, a corpus study, mora-timing, Japanese

## 1. Introduction

Recent research has shown that *informativity* can affect our speech behaviors at several linguistic levels. In phonetics, for example, it has been demonstrated that the duration of syllables and words can be influenced by how much information they carry (e.g. Aylett & Turk, 2004, 2006; Bell, Jurafsky, Fosler-Lussier, Girand, Gregory & Gildea 2003; Bell, Brenier, Gregory, Girand, & Jurafsky 2009; Cohen-Priva, 2012, 2015; Hume, 2016; Jurafsky, Bell, Gregory, Raymond 2001). More specifically, for instance, Aylett & Turk (2004) show that in English, more predictable vowels are shorter and more centralized. Bell et al. (2009) likewise show that more predictable content words are shorter in duration in conversational English. See Hall, Hume, Jaeger & Wedel (2016) for a recent, extended review of similar findings in which informativity seems to affect phonetic implementation patterns.

Shaw, Han & Ma (2014) demonstrate that informativity can play a role at the morphophonological level. In the Modern Standard Chinese truncation compounding pattern, what survives in truncation tends to be those segments that are informative, measured in terms of family size and frequency ratios between compound frequency and character frequency; i.e. those segments that best enable listeners to recover the original, untruncated words. To provide one more example, past phonological research has reported the generalization that, given a consonant cluster straddling a syllable boundary, an onset consonant never deletes; it is only the coda consonant that can delete (Wilson 2001). McCarthy (2008) develops a theory of constraint interaction which accounts for this observation, by postulating that only coda consonants can be deleted because they are targeted by CODACONDITION (Ito 1989). In a corpus study, Raymond, Dautricourt & Hume (2006) uncover exceptions to the generalization. They examined the Buckeye corpus of spontaneous interview speech, and found that onset [t] and [d] can delete in frequent words like

*somebody*, *lady*, and *better*, especially when the following context makes those words predictable (i.e. when they have high reverse transitional probabilities); e.g. [d] in *ladies and gentlemen*. Raymond et al. (2006) thus shows that even phonologically-privileged sounds like onset consonants can delete, when they are not informative. See also Cohen-Priva (2015) for a similar finding in which onset [t] can delete when it is not very informative.

A cross-linguistic study by Piantadosi, Tily & Gibson (2011) shows that informativity may affect lexical organization—they show that word lengths can be partly predicted based on informativity; more informative words tend to be longer. Seyfarth (2014) shows that words that are usually predictable tend to reduce, and that they appear as reduced even in non-predictable contexts, suggesting that the reduced forms are stored in the lexicon. Jaeger (2010) argues that informativity may affect syntactic patterns in that speakers attempt to distribute information more or less consistently across the signal. To summarize then, informativity seems to play a non-trivial role at every level of our linguistic behavior, from phonetics to syntax.

This growing body of research is evolving into a general hypothesis that speakers may articulate a linguistic unit with more information more robustly—henceforth, we call this general hypothesis the “informativity hypothesis”. The informativity hypothesis seems to be plausible from a viewpoint of efficient communication (Hall et al 2016; Hume 2016): linguistic units that carry high information should not be misperceived by the listener; on the other hand, linguistic units that can be predicted from contextual information—those with inherently low information—can be recovered by listeners, even if the signals are degraded. Indeed, as reviewed above, recent research shows that this principle may be at work in governing several aspects of our linguistic behavior.

However, there are two aspects in which this research project can and should be developed further: (i) target languages, and (ii) the targeted level of linguistic representation. First, most previous studies of this research program target Indo-European languages, including English (e.g. Arnon & Cohen-Priva 2015; Aylett & Turk 1994, 2006; Bell et al 2003, 2009; Cohen-Priva 2015), Dutch (Kuperman, Pluymaekers, Ernestus, & Baayen 2007; Pluymaekers, Ernestus & Baayen 2005; van Sol & Pols 2003), French (Bürki, Ernestus, Gendrot, Fougeron & Frauenfelder 2011; Torreira & Ernestus 2009), (Brazilian) Portuguese (Everett, Miller, Nelson, Soare & Vinson 2011), and Spanish (Cohen-Priva 2012); the only exceptions that we know of are the study of Egyptian Arabic by Cohen-Priva (2012) and the study of second-mention reduction effects in Indian English and Korean (Baker & Bradlow 2007). The principle that the signal is controlled to maximize communicative efficiency should apply in principle to any language, and thus needs to be tested in languages beyond Indo-European.

The second gap in this research is that it almost always targets syllable or word duration, and does not usually reveal interactions at segmental levels. Most work examines syllable or word duration (e.g. Arnon & Cohen-Priva 2014; Aylett & Turk 2004; Bell et al. 2003, 2009), segment duration (Bürki et al. 2011; Cohen-Priva 2015; Hanique et al. 2010; Kuperman et al. 2007; Torreira & Ernestus 2009), or a couple affixes (Pluymaekers et al. 2005) without revealing the specific phonological locus of the effects. Extending the informativity hypothesis to the domain of syntax, Jaeger (2010: 24) puts forward a strong thesis: “Human language production could be organized to be efficient at *all* levels of linguistic processing in that speakers prefer to trade off redundancy and reduction” (emphasis in the original). In order to examine this claim, we feel that it is also necessary to investigate whether the trade-off between informativity and reduction applies to the level of phonotactics, an aspect of phonological grammar. More specifically, we ask whether

vowels are reduced in contexts where their identity is more or less predictable from the preceding consonants, and, likewise, whether vowels are produced with longer duration when vowel identity is not predictable from the preceding consonant. If Jaeger's thesis is correct that the principle of efficient production applies at *all* linguistic levels, we should observe this sort of trade-off. On the other hand, it would not be surprising if we find no such effects at the local, segmental level, as the CV unit itself is generally not a meaning-bearing unit. Put differently, we ask whether informativity influences phonetic patterns at level of the phonology, where linguistic meaning is irrelevant.

With these two issues in mind, this paper assesses the informativity hypothesis by testing whether Japanese vowel duration is influenced by consonant-conditioned informativity within a CV mora. Japanese provides an interesting test case for the informativity hypothesis, since it differs rhythmically from English, and uses duration to express short vs long phonological contrast including both long vowels and geminate consonants (Han 1962, 1994; Homma 1981 *et seq.*). Japanese is also thought to be a "mora-timed" language (e.g.; Han 1962, 1994; Port, Dalby & O'Dell 1987; cf. Beckman 1982; see Warner & Arai 2001 for a critical review), such that there is some pressure in the language for mora-based isochrony. These points highlight that Japanese is different in important ways from other languages on which the informativity hypothesis has been tested. There are also specific phonetic details of Japanese that make it a particularly intriguing test case for phonologically-localized informativity. Primary amongst these is that various consonantal factors are already known to affect vowel duration in Japanese (as we will confirm below). A coarse generalization is that the same vowel tends to be produced longer after a phonetically shorter consonant than after a longer consonant. This observation has been taken as evidence that Japanese speakers keep the duration of CV units more or less constant (e.g. Homma

1980; Port, Al-Ani & Maeda 1980; Sagisaka & Tohkura 1984). Port et al. (1987) found a strong linear correlation between word duration and the number of moras that the word contains (see also Han 1994 for further support and Arai and Warner 2001 for critique). What is most interesting about the above facts in connection with the current study is that Japanese is a language in which there is substantial variation in vowel duration within the CV unit that appear to be conditioned in some way by properties of the preceding consonantal environment.

To summarize, the current study computes informativity as the conditional entropy of a vowel given the preceding consonant and tests whether informativity affects vowel duration. This study also examines various other factors—vowel quality, preceding consonantal features, syllable structure, and others—which have been previously found to affect vowel duration in Japanese. This analysis addresses if and how consonant-conditioned entropy affects a vowel's duration beyond these potentially confounding factors.

## **2. Method**

### **2.1. The speech corpus**

The analysis is based on the Corpus of Spontaneous Japanese (the CSJ: Maekawa, Koiso, Furui, & Ishihara 2000), one of the largest annotated speech corpora of Japanese. The CSJ contains several speech styles, including, but not limited to, Academic Presentation Style (APS) and Spontaneous Presentation Style (SPS), the former of which is based on real academic speech, which is more formal. The latter is solicited speech recorded at the recording room in the National Institute for Japanese Language and Linguistics (NINJAL). The speakers were given a topic about their life, e.g., “tell us your happiest moment in life?” as a prompt. The speech was monologue,

but there are 3 to 4 listeners at the time of recording. The speakers' age range was mainly 30's to 70's. The gender was more or less balanced, although they were slightly more male speakers.<sup>1</sup>

The current analysis used the core portion of the corpus (known as the CSJ-RDB), which comes with rich annotation and phonetic information. The CSJ-RDB consists of 11,559 unique words produced by 70 speakers, and over 312,000 vowel tokens. The CSJ-RDB includes annotated segmental intervals, created by hand, rather than using some sort of forced aligner.<sup>2</sup>

## **2.2. Consonant-conditioned vowel entropy**

The studies cited and reviewed in section 1 have used several different measures to quantify informativity, and it is safe to say that we are still exploring what the right measure of informativity is, to the extent that informativity plays a role in linguistic patterning at all, for any given dataset. For example, Cohen-Priva (2015) used four information theoretic measures to predict segment duration in American English: word frequency, segment probability, segment informativity, and segment predictability. As our purpose in this paper is to focus on the possibility that informativity plays a role within the phonology, we set aside word frequency. Segment probability is unlikely to have an effect on Japanese vowel duration since the most frequent vowel, /a/, is also the longest (see section 3, Table 1 for the actual data). In Cohen-Priva (2015), segment predictability captured

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<sup>1</sup> For further details, which should not be relevant for the current analyses (such as the whole list of topics in APS and recording equipment), see the documentation available at [http://pj.ninjal.ac.jp/corpus\\_center/csj/manu-f/recording.pdf](http://pj.ninjal.ac.jp/corpus_center/csj/manu-f/recording.pdf).

<sup>2</sup> Thanks to Hanae Koiso (p.c.) for answering some of our questions regarding the CSJ-RDB. An anonymous reviewer asked how the CSJ-RDB determined a boundary between [w] and [a]. According to the manual available at [http://pj.ninjal.ac.jp/corpus\\_center/csj/k-report-f/06.pdf](http://pj.ninjal.ac.jp/corpus_center/csj/k-report-f/06.pdf), they first (i) determined the end of the steady state of the preceding vowel, then (ii) determined the midpoint of the glide (located based on the formant peak), and (iii) the onset of the following vowel. The end of the glide was marked at the middle point between (ii) and (iii). Devoiced vowels were often not distinguishable from the preceding consonants, and hence often merged with them, ending up not having its own interval.

local effects of context, while segment informativity (average predictability across contexts) captured the general tendency of a segment to be longer or shorter, independently of the local context. The informativity measure thus captures a more abstract property of the segment albeit one that owes its computation to the lexical statistics of the language. The facts of Japanese vowel duration have directed us towards a similarly abstract measure of informativity but one that abstracts over phonotactic environments as opposed to individual segments.

Following other recent work (e.g. Cohen-Priva 2012, 2015; Daland, Oh & Kim 2015; Hall 2009; Hume 2016; Hume et al. 2016; Kawahara 2016b), the current study made crucial use of Shannon's (1948) entropy to quantify informativity. Vowel entropy is defined as the weighted average of the surprisal of each vowel. The surprisal term is the negative log of a vowel's probability:  $-\log_2 p(x)$ . The surprisal term is multiplied by the un-transformed probability of the vowel,  $p(x)$ , which serves as the weight. To capture how the vowel informativity is influenced by the preceding consonantal context,  $H(V)$  was calculated over the five Japanese vowels, /a/, /e/, /i/, /o/, /u/, in each consonantal environment in the corpus:  $(H(V) = - \sum_{x \in V} p(x) * \log_2 p(x))$ . This measure provides in the domain of phonotactics a measure that is the conceptual equivalent of segmental informativity in Cohen-Priva (2015). While Cohen-Priva (2015) computed a measure of segmental informativity that averaged across phonological contexts, we computed a measure of phonotactic informativity that averages across segments (vowels, in our case). The result is a single measure that quantifies vowel uncertainty in a given consonantal context. We refer to this measure of phonotactic informativity as CVEntropy for "Consonant-conditioned Vowel Entropy". The higher the CVEntropy, the less predictable/more informative that vowel is in the specified consonantal context.



There are a few advantages of using this particular measure to quantify informativity. One is that it is based on Shannon's entropy which is very simple to calculate and, defined within the larger framework of Information Theory (Shannon 1948, *et seq.*),<sup>3</sup> and therefore allows us to relate the current work within the overall research enterprise using Information Theory, in linguistics and beyond. More importantly, CVEntropy offers a direct measure of how much information a vowel carries given a particular consonantal context. It captures a property of phonological environments that is abstracted over lexical statistics. Because it is abstract, it can remain stable even as local predictability changes, the phonotactic analogue of segmental informativity.

It is in large part the facts of Japanese vowels that have led us to pursue phonotactic informativity as opposed to segmental informativity. Japanese has phonotactic restrictions—gradient and categorical—that reduce the number of vowels that can follow certain consonants and this influences informativity, as quantified with CVEntropy. For example, since front vowels are prohibited after palatalized consonants, it is easier to predict vowel quality, either /a/, /u/, or /o/, in these environments; i.e., in these cases, CVEntropy is low. On the other hand, the distribution of the five vowels can be unpredictable given a preceding consonant, in which case the vowel is informative and, accordingly, its CVEntropy is high. As we will observe (Figure 1 below), the

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<sup>3</sup> Daland et al. (2015) proposed to make use of Shannon's entropy to explore the contributions of orthography and speech perception in the context of loanword adaptation. In this sense, entropy is a tool that is independently shown to be useful in linguistic exploration. We do not mean, however, that Shannon's entropy *must* be the right tool for this sort of analysis; much more work needs to be conducted in order to address what measure of informativity is the right tool; and it may be the case that different measures may be helpful to model effects of informativity at different levels. As Robert Daland puts it in his review “we are still exploring the space of informativity measures”, and we agree with this statement. The current exploration should be taken as one case study, trying out the effectiveness of Shannon's entropy to explore the CV-interaction. See also section 4 for further discussion.

degree of variability in CVEntropy across consonantal environments is sufficient to quantitatively access the effects of CVEntropy on vowel duration.

On the other hand, segmental informativity would only function to separate the front vowels, /i/ (5.68 bits) and /e/ (5.27 bits), which tend to have low informativity, from the other vowels, /a/ (11.04 bits), /u/ (10.20 bits) and /o/ (10.20 bits), which have higher informativity, defined here as average entropy across consonantal contexts in the CSJ corpus. This measure would not be very helpful for two reasons. First, this (almost) dichotomous distinction would not distinguish the five vowels in Japanese very well. Second, empirically speaking, this division does not pick out duration differences. From the previous work on vowel duration in Japanese (Campbell 1992, 1999; Han 1962; Sagisaka & Tohkura 1984), we know that of the high informativity vowels, /a/ tends to have long phonetic duration, while /u/ is the shortest vowel (see also Table 1). Similarly for the low informativity vowels, /e/ tends to be long while /i/ tends to be short. Phonotactic informativity, or CVEntropy, thus offers a better alternative, providing an abstract characterization of the context (as opposed to the segment) that conditions variation in duration.

One of the intuitions behind informativity effects more broadly is that greater uncertainty corresponds to increased competition in speech production which leads to longer word durations (Bell et al., 2009; Kuperman & Bresnan 2012). There is a body of evidence supporting cascading activation in speech production (Goldrick & Blumstein 2006; Mcmillan & Corley 2010). Put simply, as one gesture is being produced, the next is being planned. The simultaneity of planning and production leads to interactions that can be observed in both naming latencies (the time required to initiate production of a word) and the resulting phonetics (Baese-Berk & Goldrick 2009; Shaw, 2013). In extreme cases, two gestures can be produced simultaneously, resulting in

speech errors to varying degree (Goldstein, Pouplier, Chen, Saltzman & Byrd 2007; Mcmillan & Corley 2010; Mowry & McKay, 1990). In cognitive models of speech production, the time required to resolve competition is a function of various language-specific parameters, including the degree of competition in a given environment (Dell, 1986; Roon & Gafos, 2016; Tilsen, 2014). The time required to initiate production (i.e., naming latency) and produce a word (i.e., word duration) both increase with uncertainty about phonological form (Shaw, 2012). Given these considerations, the informativity hypothesis predicts a positive correlation between CVEntropy and vowel duration.

Our measure of phonotactic informativity is well-suited to evaluate the hypothesis that contextually-determined uncertainty is phonologized such that average vowel predictability in a given context regulates vowel duration. CVEntropy values were calculated based on the conditional probabilities of vowels given preceding consonants in the CSJ-RDB corpus. In keeping with the aim of this paper to explore informativity within the domain of phonotactics, we calculated CVEntropy based on type frequencies (as opposed to token frequencies) in the corpus. Duration values are based on those provided in the CSJ-RDB. We examined the correlations between CVEntropy and vowel duration, as well as other factors that have been claimed to affect vowel duration. The primary question is whether CVEntropy helps to explain vowel duration patterns, beyond those effects that are already known to affect vowel duration. Along these lines, we also ask whether these effects that were previously known to affect vowel duration may possibly be explained instead by CVEntropy.

### **3. Results**

#### **3.1. CVEntropy by preceding consonant environment**

Figure 1 shows how the CVEntropy varies across consonantal environments. We have excluded consonants that are under-represented in the corpus, showing only consonant environments with at least 1,000 occurrences in the corpus.<sup>4</sup> The vertical axis represents CVEntropy. Consonant environments, shown on the horizontal axis, are ordered from low to high entropy. The theoretical maximum of CVEntropy given 5 vowels is 2.32 ( $-\log_2 p(0.2)$ ), which happens when all 5 vowels appear with the same probability ( $1/5=0.2$ ). The solid black line indicates the CVEntropy of the vowel in each consonantal environment in Japanese. The consonantal environment that conditions the highest vowel entropy is /m/, which is close to the theoretical maximum. There are several other consonants, e.g., /h/, /r/, /t/, /k/, /g/, /s/, with comparably high CVEntropy. At the left side of figure, we find the consonant environments that condition low CVEntropy. The consonant environment with the lowest CVEntropy, /w/, is almost always followed by /a/, except in some loanwords like [wisukii] ‘whisky’. Thus, /w/, is a near perfect predictor of following vowel quality. Since the vowel following /w/ is highly predictable, it carries little information content, and its CVEntropy is near zero. In between low entropy /w/ and the group of high entropy consonants there is a roughly linear increase across the various palatal consonants, /hy/, /sy/, /y/, /zy/, and then voiced coronals, /d/, /n/, /z/, and /b/.

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<sup>4</sup> Consonants that occurred less than 1,000 times were: /dy/, /kw/, /ty/, /ny/, /v/, /ry/, /ky/, /cy/, /py/, /by/, /my/, and /p/.

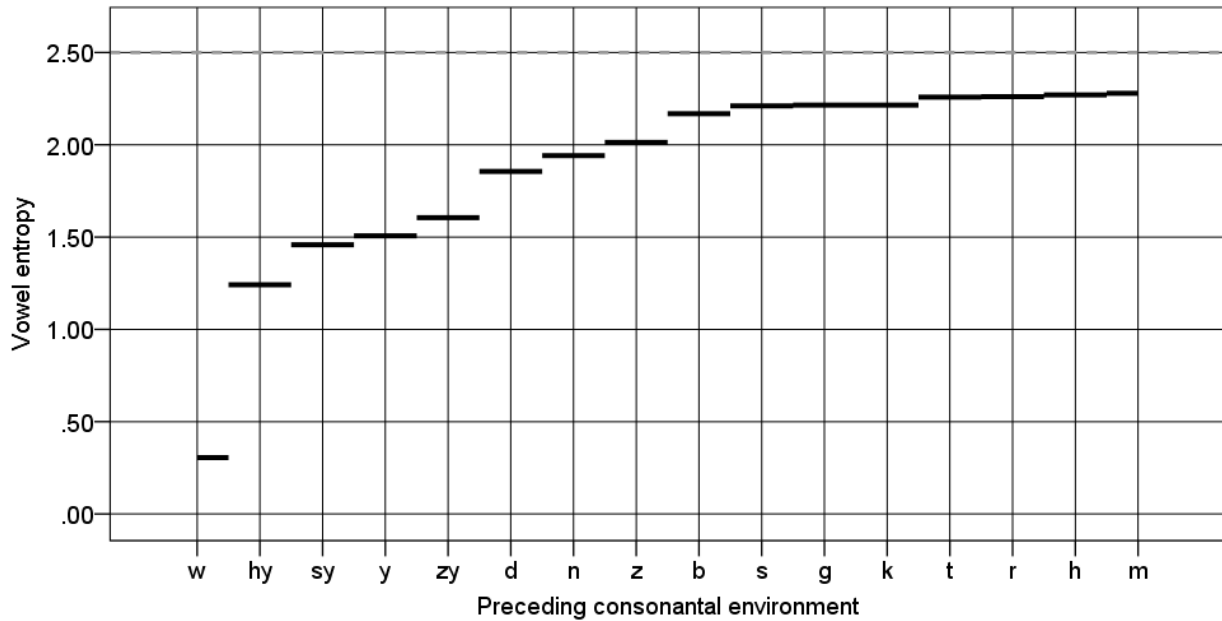


Figure 1: The CVEntropy (Consonant-conditioned Vowel Entropy), ordered from low to high. [Xy] represents a palatalized version of X, the convention used in the CSJ. /hy/ is phonetically realized as [ç], /sy/ as [ɕ] and /zy/ as [ʑ]. See Vance (2008).

Overall, Figure 1 indicates that there is substantial range in CVEntropy as a function of the preceding consonant environment. This variation allows us to assess whether CVEntropy affects vowel duration.

### 3.2. Vowel duration for each vowel

Figure 2 shows the distribution of vowel duration for each of the five Japanese vowels in the corpus. There were 361,241 vowel tokens in the portion of the corpus analyzed. For the current analysis, phonemically long vowels were excluded (n=44,786), because their frequencies are incomparably lower than those of short vowels, as were phonemically short vowels that were extreme outliers

( $\pm 3$  SD from mean) in duration ( $n=5,357$ ). We also excluded vowels that followed low frequency consonants, those that occurred fewer than 1,000 times in the corpus ( $n = 47,684$ ). After these exclusions, 263,414 tokens remained in the analysis. The shape of the distributions for each of the 5 vowels is similar: all have long right tails and steeper left tails that fall towards zero.

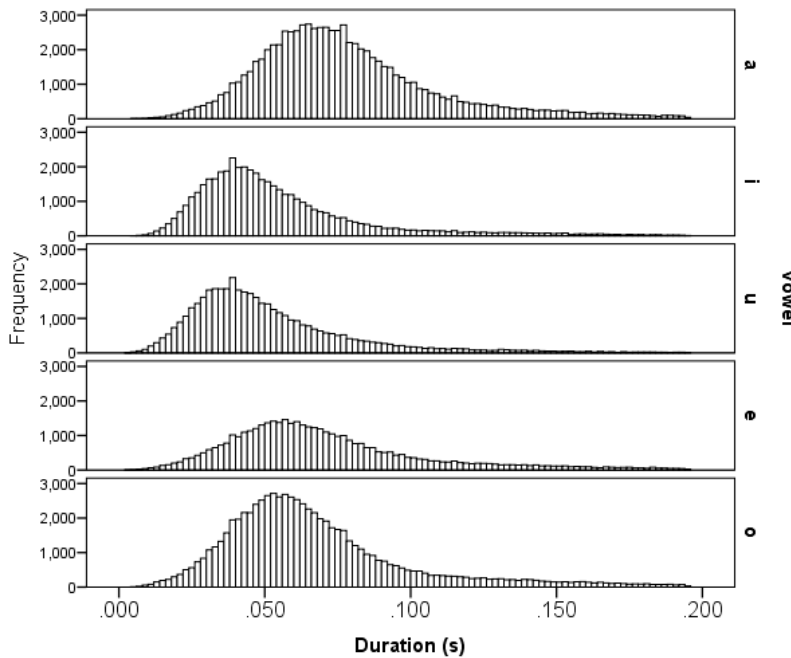


Figure 2: The distribution of vowel duration for each vowel.

|       | Mean | SD | N       |
|-------|------|----|---------|
| /a/   | 78   | 30 | 77,729  |
| /e/   | 70   | 33 | 39,051  |
| /o/   | 67   | 31 | 62,094  |
| /i/   | 54   | 29 | 44,070  |
| /u/   | 52   | 29 | 40,470  |
| total | 66   | 32 | 263,414 |

Table 1. The number of valid token counts along with the mean and SD in ms. of the five vowels in Japanese.

Table 1 provides descriptive statistics for vowel duration by vowel. The mean duration of the five vowels follow the order of /a/ > /e/ > /o/ > /i/ > /u/, which is compatible with what is found in the previous studies on Japanese vowel duration (Arai, Warner & Greenberg 2001; Campbell 1992, 1999; Han 1962; Sagisaka 1985; Sagisaka & Tohkura 1984)—we take this replication as evidence that our data source, the CSJ-RDB, is reliable. The SD of vowel duration is rather high. For the high vowels, /i/ and /u/, the standard deviation is greater than half the mean.

### 3.3. Vowel duration in different consonantal environments

Figure 3 illustrates, for each vowel, how vowel duration (y-axis) changes as CVEntropy (x-axis) increases. For reference, the gray line which shows the pattern for [a] is superimposed on the other panels. The consonantal environments on the x-axis are ordered from low (left) to high (right) CVEntropy.

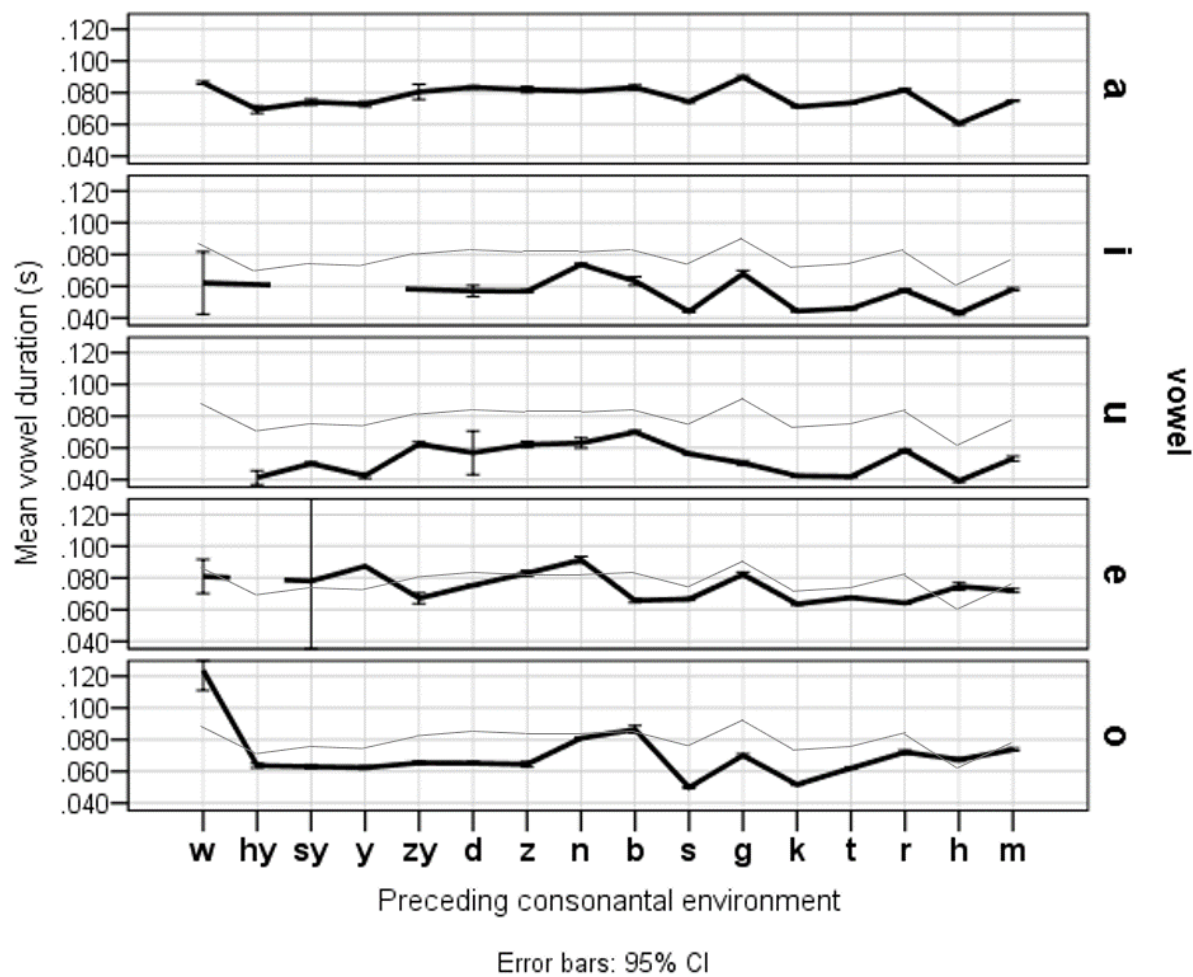


Figure 3: Vowel duration after different preceding consonants, broken up by vowel types. The consonants are ordered from low (left) to high (right) CVEntropy. For reference, the grey line shows the pattern for /a/ superimposed on the other vowels.

At first sight, there may not seem to be a straightforward correlation between vowel duration and CVEntropy. However, upon careful examination, we observe other factors affecting vowel durations in Figure 3. For example, vowels are longer after voiced stops than after voiceless stops (compare /t/ vs. /d/ and /k/ vs. /g/ in Figure 3).<sup>5</sup> This effect of voicing on the following vowel is illustrated in Figure 4. The voicing effect has been found in lab speech obtained in previous production experiments, and a previously given explanation is that since voiced stops are shorter, the following vowels are longer due to mora-timing (Port et al. 1980; Sagisaka & Tohkura 1984). We also observe an effect of place of articulation. Compare for example /m/ and /b/ on the one hand, and /k/ on the other in Figure 3; it seems that vowels tend to be longer when following labial consonants than when followed by dorsal consonants. The effect of place of articulation is shown in Figure 5. It actually shows that a vowel that is preceded by a more front consonant is longer.

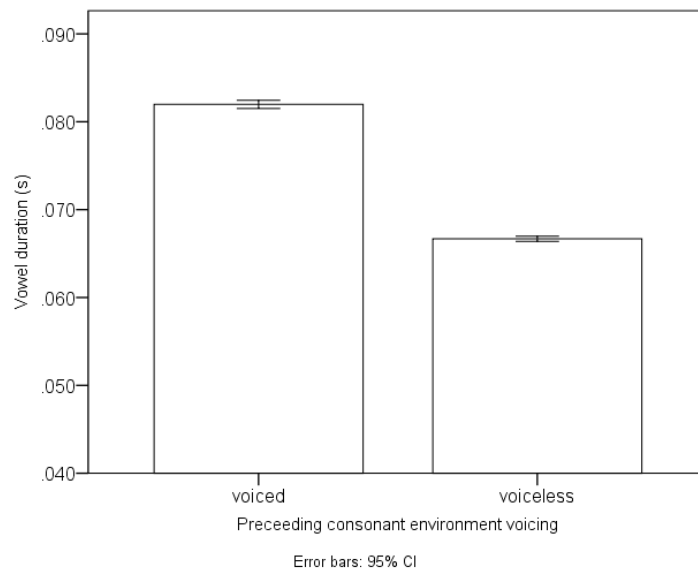


Figure 4: The average vowel duration after voiced (including both voiced obstruents and sonorants) and voiceless consonants.

<sup>5</sup> Japanese has lost /p/ in its history, and therefore (singleton) /p/ only appears in loanwords and is thus rare in the overall Japanese lexicon (Ito & Mester 1995). This is why /p/ does not enter into the current analysis.



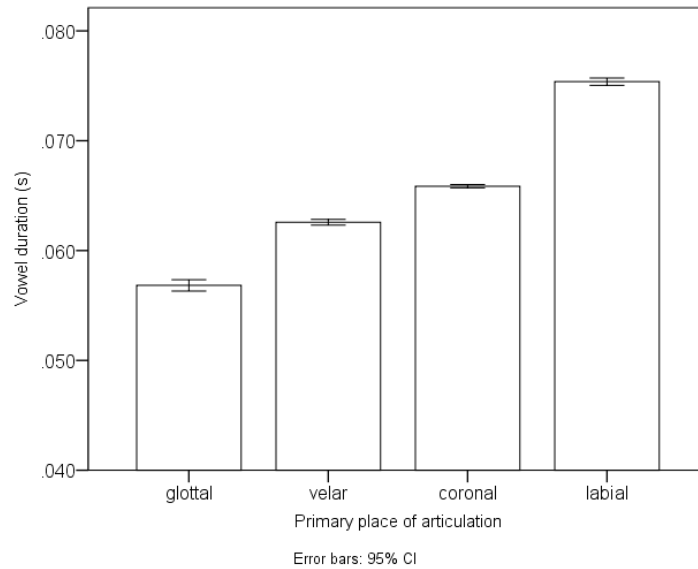


Figure 5. The average vowel duration after consonants with different primary place of articulation.

Going back to Figure 3, the effect of place of articulation, however, is not uniform across vowels. One tendency is that vowels that share an articulator with the following consonant are longer. Comparing vowels following /g/ vs. /b/, /u/ and /o/—vowels involving some control of the lips—are longer after /b/ than after /g/; /a/ and /e/—vowels involving the positioning of the tongue body—are longer after /g/ than after /b/; in contrast, /i/, which involves palatal approximation by the tongue blade, is similar across /b/ and /g/. Thus, there are multiple interactions between vowel identity and place of articulation of the preceding consonant.

Across the entire corpus, /a/ is, on average, longer than the other vowels, but the magnitude of this difference is conditioned by consonantal context. There are even some consonantal environments in which /a/ is shorter than /e/ and /o/. To highlight this, in Figure 3, we have superimposed the pattern for /a/ across consonants on the panels showing the other vowels, /i/, /u/, /e/, /o/. When following /n/ and /h/, /e/ is longer than /a/; following /b/, /o/ is longer than /a/. Overall

these observations imply that there are numerous phonetic effects that may obscure the influence of CVEntropy on vowel duration.

Nevertheless, it is still possible to identify some trends of CVEntropy in the predicted direction in Figure 3. For example, the duration of /a/ gradually increases with CVEntropy from the /hy/ context to the /b/ context. Recall from Figure 1 that this is the range of CVEntropy over which we see interesting variation. The general trend for /u/, as well, is for a gradual increase in duration across this range, although /e/ and /o/ do not follow suit. At the least, Figure 3-5 indicate that there may be some promise in CVEntropy but revealing it requires that we control for numerous other factors.

In addition to those effects examined in Figures 3-5, another factor known to influence vowel duration is syllable structure. Japanese has closed syllables, where the coda consonants are limited to a so called “coda-nasal” (Vance 2008) or the first part of a geminate (Kawahara 2016a; Vance 2008). Figure 6 illustrates the durations of vowels in open and closed syllables. As shown in previous production studies, Japanese vowels are longer in closed syllables than in open syllables (Campbell 1999; Han 1994; Idemaru & Guion 2008; Kawahara 2006; Port et al. 1987).

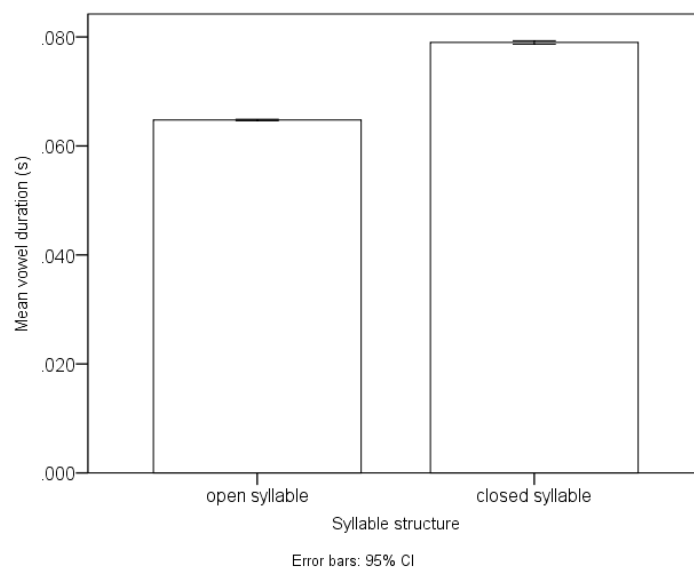


Figure 6: The effects of syllable structure on vowel duration.

The above observations (Figures 3-6) show that, in order to evaluate the effect of CVEntropy on duration, we need to take other effects into account. To that end, we fit two generalized linear models to the data. One is the baseline model, which involves factors that condition following vowel duration, including those presented above. The other one adds CVEntropy as an additional predictor. A comparison between these two models allows us to assess the effect of vowel entropy in the presence of other factors that are known to influence vowel duration.

### **3.4. The model comparison**

The baseline model contained the following fixed factors: VOWEL quality (a, i, u, e, o), VOICING (voiced vs. voiceless), primary PLACE of articulation (glottal, coronal, labial, velar), SONORANCY (sonorant vs. obstruent), and SYLLABLE STRUCTURE (open vs. closed syllables). The fixed factors of VOWEL, VOICING, PLACE, SONORANCY, and SYLLABLE STRUC(TURE) were dummy coded with the first level as the reference category: /a/ for VOWEL; voiced consonants for VOICING; glottal consonants, /h/ and /hy/, for PLACE of articulation; sonorants, /w/, /y/, /n/, /r/, /m/, for the SONORANCY factor; and, open syllables for SYLLABLE STRUC.

All interactions between VOWEL quality and the other fixed factors were also included in the baseline model. Random intercepts for talker and for word and random slopes varying with CVEntropy were also included in the baseline model (the last of which is necessary for model comparison). Table 2 provides a summary of the fixed factors in the baseline model; Table 3

summarizes the CVEntropy model. Both models were fit to 263,414 data points (see the method section).

Table 2: Baseline model: duration ~ vowel\*voicing + vowel\*place + vowel \*sonorancy+  
vowel\*syllable\_struc +(1+CVEntropy|talker)+(1|word)

|                        | Estimate | Std. Error | t value |
|------------------------|----------|------------|---------|
| (Intercept)            | 0.064    | 0.0011     | 58.64   |
| VOWEL_e                | 0.007    | 0.0021     | 3.24    |
| VOWEL_i                | -0.006   | 0.0013     | -4.24   |
| VOWEL_o                | 0.014    | 0.0014     | 9.84    |
| VOWEL_u                | 0.001    | 0.0170     | 0.06    |
| VOICING                | -0.010   | 0.0005     | -21.57  |
| VELAR                  | 0.009    | 0.0008     | 11.75   |
| CORONAL                | 0.015    | 0.0008     | 20.02   |
| LABIAL                 | 0.004    | 0.0009     | 5.00    |
| SONORANCY              | 0.005    | 0.0005     | 9.04    |
| SYLLABLE_STRUC         | 0.015    | 0.0005     | 33.96   |
| VOWEL_e:VOICING        | 0.009    | 0.0008     | 11.07   |
| VOWEL_i:VOICING        | -0.001   | 0.0008     | -1.48   |
| VOWEL_o:VOICING        | -0.004   | 0.0007     | -5.18   |
| VOWEL_u:VOICING        | 0.004    | 0.0008     | 4.78    |
| VOWEL_e:VELAR          | -0.008   | 0.0020     | -4.10   |
| VOWEL_i:VELAR          | 0.002    | 0.0013     | 1.62    |
| VOWEL_o:VELAR          | -0.018   | 0.0013     | -13.39  |
| VOWEL_u:VELAR          | -0.022   | 0.0170     | -1.27   |
| VOWEL_e:CORONAL        | -0.016   | 0.0020     | -7.92   |
| VOWEL_i:CORONAL        | -0.010   | 0.0012     | -7.82   |
| VOWEL_o:CORONAL        | -0.019   | 0.0013     | -14.81  |
| VOWEL_u:CORONAL        | -0.021   | 0.0170     | -1.22   |
| VOWEL_e:LABIAL         | 0.001    | 0.0022     | 0.32    |
| VOWEL_i:LABIAL         | -0.001   | 0.0016     | -0.47   |
| VOWEL_o:LABIAL         | -0.016   | 0.0015     | -10.38  |
| VOWEL_u:LABIAL         | -0.011   | 0.0170     | -0.65   |
| VOWEL_e:SONORANCY      | -0.010   | 0.0008     | -12.26  |
| VOWEL_i:SONORANCY      | -0.014   | 0.0009     | -15.73  |
| VOWEL_o:SONORANCY      | -0.008   | 0.0008     | -9.36   |
| VOWEL_u:SONORANCY      | -0.007   | 0.0009     | -8.16   |
| VOWEL_e:SYLLABLE_STRUC | 0.001    | 0.0007     | 1.04    |
| VOWEL_i:SYLLABLE_STRUC | -0.004   | 0.0009     | -4.56   |
| VOWEL_o:SYLLABLE_STRUC | 0.003    | 0.0008     | 3.15    |
| VOWEL_u:SYLLABLE_STRUC | -0.001   | 0.0009     | -1.70   |

We start with a description of the baseline model. The intercept represents an abstract reference category (/a/ before consonants that are voiced, glottal, non-palatal, sonorant). At 64 ms, the intercept is very near the average vowel duration in the data, 66 ms (Table 1). The fixed factors show a mix of negative and positive effects, explaining the substantial variation around the mean. Somewhat surprisingly, two of the vowels, /e/ and /o/, have significant positive coefficients, indicating that—once other factors are taken into account—these vowels have a longer “intrinsic” duration than the baseline vowel /a/. The other fixed factors, VOICING, PLACE, SONORANCY, and SYLLABLE STRUC, also had strong effects and each of these factors interacted with one or more vowels. The significant interactions indicate that these effects were never completely uniform across vowels. A few factors, VOICING and SYLLABLE STRUC pattern in the same direction across vowels, even though the magnitude of the effect differs. The main effect of VOICING was relatively large (10 ms) and negative, indicating that vowels were shorter following voiceless stops than voiced stops. Inspection of the VOWEL\*VOICING interaction terms reveals that some vowels showed greater effects than others. The effect of voicing on /o/ was even greater than the main effect while the effect of voicing on /e/ and /u/ was attenuated. Despite these interactions, the sum of the main effect of VOWEL and the VOWEL\*VOICING interaction was negative for all vowels. Thus, the baseline model indicates a negative effect of voicing that differs in degree by vowel. The case is similar for SYLLABLE STRUC in that there is a large (15 ms) positive effect and some smaller (1~4 ms) interactions with vowels. The direction of the effect indicates that vowels are significantly longer in closed syllables than in open syllables, a trend that is consistent with other work on Japanese (see Figure 6 above) but which bucks the cross-linguistic tendency (Maddieson, 1985). PLACE of Articulation and SONORANCY also have strong and reliable effects, but the pattern of interaction with vowels is more complex. For PLACE (GLOTTAL, VELAR, CORONAL, LABIAL), the

main effects are positive in part because /a/, the baseline vowel, is particularly short before glottals, the baseline PLACE of articulation. The situation is similar for SONORANCY. The baseline vowel /a/ is longer after obstruents than after sonorants but the other vowels show the opposite effect. We return to this discussion in the context of vowel-specific models later in the section. Overall, the baseline model provides estimates for fixed factors that are reasonable given the descriptions of the data above (e.g., Figures 2- 6).

Incorporating CVENTROPY and the interaction between CVENTROPY and VOWEL results in significant improvement over the baseline model (Table 3). Table 4 summarizes the model comparison. The lower Akaike Information Criterion (AIC) for the ENTROPY model indicates that the additional model complexity of this model resulting from inclusion of CVENTROPY and CVENTROPY\*VOWEL factors is justified by the increased log likelihood of the data. Both CVENTROPY and the interaction between CVENTROPY and VOWEL are significant predictors in the model. The presence of these factors also influences the estimates of other fixed factors, most notably: VOWEL and some of the interaction terms including VOWEL. The strong positive effects of /o/ and /e/ on duration observed in the baseline model have given way to weak negative effects. Thus, with CVENTROPY in the model, /a/ emerges as having the longest duration, which we expect given the descriptions of the data. It is also relevant to note in this context that the intercept estimate has also shifted up, from 64 ms in the baseline model to 67 ms in the CVEntropy model, which is even closer to the average vowel duration in the corpus under analysis. The changes in estimate occur because CVENTROPY explains some of the variance attributed to the VOWEL factor in the baseline model. The effect of CVENTROPY in the model is complex. It has a small (1 ms) but reliable main effect and much larger interactions with vowel. We return to the by-vowel effect of CVENTROPY after evaluating some of the control variables in the model.

Table 3: Entropy model: duration ~ vowel\*voicing + vowel\*place + vowel

\*sonorancy+vowel\*CVEntropy+ vowel\*syllable\_struc +(1+CVEntropy|talker)+(1|word)

|                        | Estimate | Std. Error | t value |
|------------------------|----------|------------|---------|
| (Intercept)            | 0.067    | 0.0018     | 36.55   |
| VOWEL_e                | -0.025   | 0.0056     | -4.52   |
| VOWEL_i                | -0.046   | 0.0079     | -5.85   |
| VOWEL_o                | -0.004   | 0.0027     | -1.30   |
| VOWEL_u                | -0.002   | 0.0172     | -0.10   |
| VOICING                | -0.010   | 0.0005     | -21.45  |
| VELAR                  | 0.009    | 0.0008     | 11.80   |
| CORONAL                | 0.015    | 0.0008     | 20.00   |
| LABIAL                 | 0.004    | 0.0009     | 4.29    |
| SONORANT               | 0.005    | 0.0005     | 9.08    |
| SYLLABLE_STRUC         | 0.016    | 0.0005     | 34.00   |
| CVENTROPY              | -0.001   | 0.0005     | -2.21   |
| VOWEL_e:VOICING        | 0.005    | 0.0010     | 5.41    |
| VOWEL_i:VOICING        | -0.004   | 0.0010     | -4.00   |
| VOWEL_o:VOICING        | -0.005   | 0.0008     | -6.74   |
| VOWEL_u:VOICING        | 0.004    | 0.0008     | 4.48    |
| VOWEL_e:VELAR          | -0.008   | 0.0020     | -4.18   |
| VOWEL_i:VELAR          | 0.003    | 0.0013     | 2.07    |
| VOWEL_o:VELAR          | -0.017   | 0.0013     | -12.48  |
| VOWEL_u:VELAR          | -0.022   | 0.0171     | -1.27   |
| VOWEL_e:CORONAL        | -0.015   | 0.0020     | -7.55   |
| VOWEL_i:CORONAL        | -0.009   | 0.0012     | -7.14   |
| VOWEL_o:CORONAL        | -0.018   | 0.0013     | -13.53  |
| VOWEL_u:CORONAL        | -0.021   | 0.0171     | -1.23   |
| VOWEL_e:LABIAL         | 0.001    | 0.0022     | 0.34    |
| VOWEL_i:LABIAL         | -0.001   | 0.0016     | -0.34   |
| VOWEL_o:LABIAL         | -0.015   | 0.0015     | -10.04  |
| VOWEL_u:LABIAL         | -0.011   | 0.0171     | -0.62   |
| VOWEL_e:SONORANT       | -0.007   | 0.0010     | -6.46   |
| VOWEL_i:SONORANT       | -0.011   | 0.0010     | -11.00  |
| VOWEL_o:SONORANT       | -0.008   | 0.0008     | -9.01   |
| VOWEL_u:SONORANT       | -0.007   | 0.0009     | -7.97   |
| VOWEL_e:SYLLABLE_STRUC | 0.000    | 0.0007     | 0.60    |
| VOWEL_i:SYLLABLE_STRUC | -0.003   | 0.0009     | -3.89   |
| VOWEL_o:SYLLABLE_STRUC | 0.003    | 0.0008     | 3.39    |
| VOWEL_u:SYLLABLE_STRUC | -0.002   | 0.0009     | -1.72   |
| VOWEL_e:CVENTROPY      | 0.014    | 0.0023     | 6.24    |
| VOWEL_i:CVENTROPY      | 0.018    | 0.0034     | 5.24    |

|                          |       |        |      |
|--------------------------|-------|--------|------|
| <b>VOWEL_o:CVENTROPY</b> | 0.008 | 0.0011 | 7.20 |
| <b>VOWEL_u:CVENTROPY</b> | 0.001 | 0.0015 | 0.93 |

Table 4: The model comparison summary

| <b>Fixed Factor</b>    | <b>Df</b> | <b>AIC</b> | <b>BIC</b> | <b>logLik</b> | <b>deviance</b> | <b>Chisq</b> | <b>Df</b> | <b>Pr(&gt;Chisq)</b> |
|------------------------|-----------|------------|------------|---------------|-----------------|--------------|-----------|----------------------|
| <b>baseline</b>        | 40        | -1151931   | -1151512   | 576006        | -1152011        |              |           |                      |
| <b>CVEntropy model</b> | 45        | -1152033   | -1151561   | 576061        | -1152123        | 111.36       | 5         | < 2.2e-16            |

To visualize the interactions between VOWEL and other factors in the CVENTROPY model, Figures 7-10 show least square means (using “lsmeans” R package: Lenth 2016), also known as “predicted marginal means”, for VOICING, SYLLABLE STRUC, SONORANCY and PLACE by VOWEL. The lsmeans are model predictions for factors of interest that take into account other model estimates. The full model, summarized in Table 3, provides the reference for the predicted values. Error bars indicate 95% confidence intervals.

Figure 7 shows that the effect of VOICING is consistent in direction across vowels—vowels are shorter following voiceless consonants than voiced consonants—but that the magnitude of the effect varies across vowels. Especially, the magnitude of the error bars for /u/ stands out. The standard error of the estimate for /u/ is substantially higher than for any other vowel. Figure 8 shows lsmeans for SYLLABLE STRUC. All five vowels are longer in closed syllables than in open syllables, and, again, /u/ is particularly variable. SONORANCY and PLACE show more complicated interactions. Vowels tend to be longer after SONORANTS than after OBSTRUENTS, but /a/ is an exception and shows the reverse pattern. The PLACE of articulation results show a detailed pattern of interaction. When other factors are controlled for, the main effect of PLACE shown in Figure 5 does not hold across vowels nor does the observation made of Figure 3 that shared place between



the consonant leads toward longer vowel duration. Rather, against the reference of the CVENTROPY model, we see a pattern of PLACE effects on vowels that is different for each vowel.

Figure 7: lsmeans for voicing by vowel. Error bars indicate 95% confidence intervals.

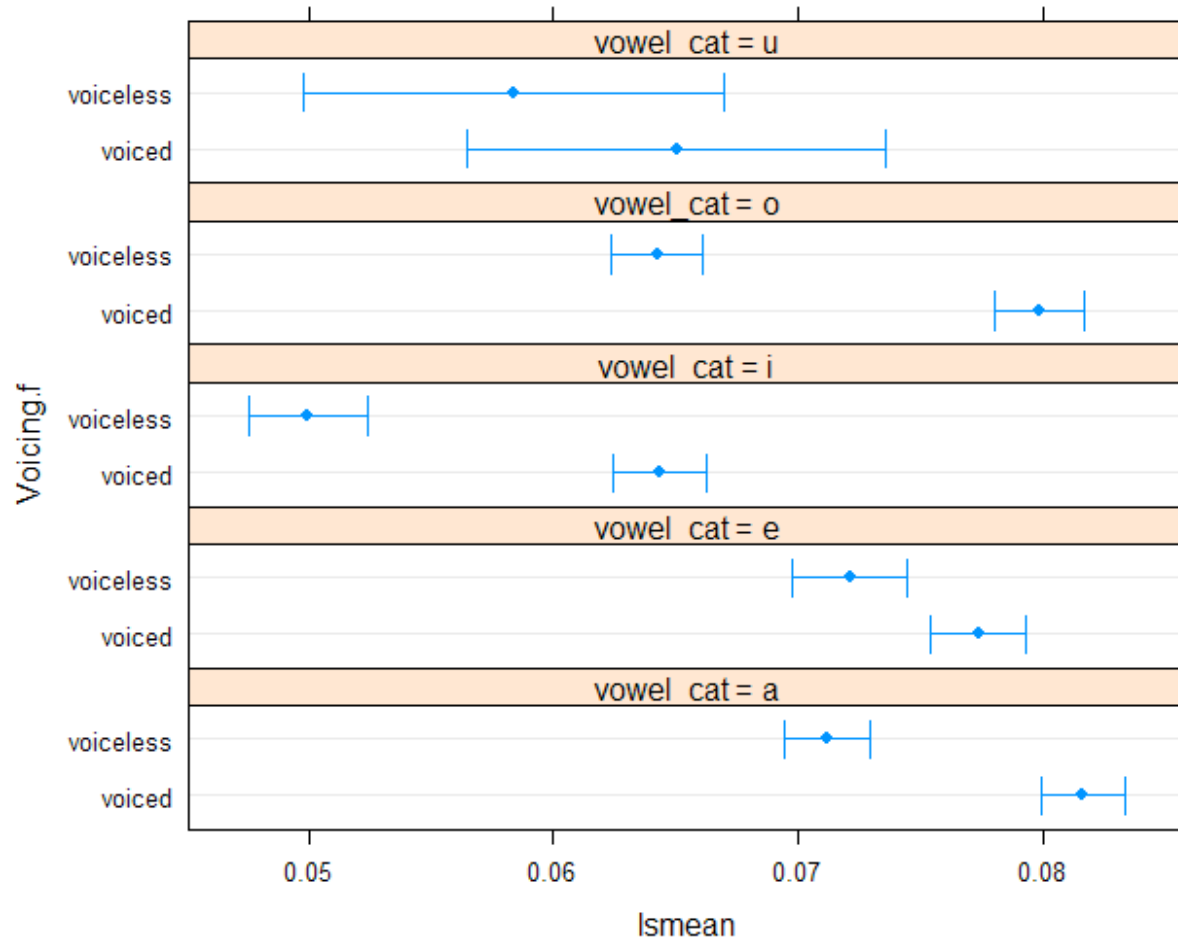


Figure 8: lsmeans for syllable structure by vowel. Error bars indicate 95% confidence intervals.

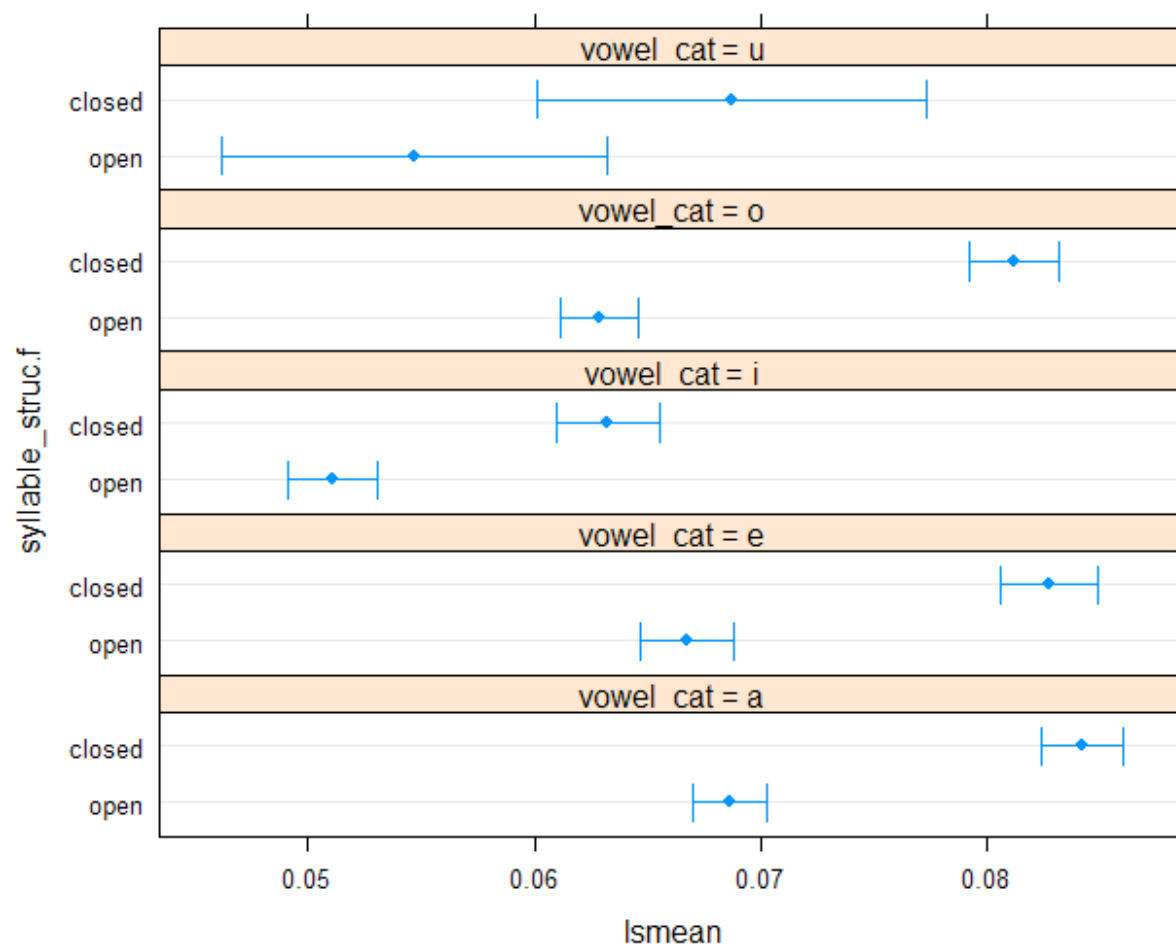


Figure 9: lsmeans for sonorancy by vowel. Error bars indicate 95% confidence intervals.

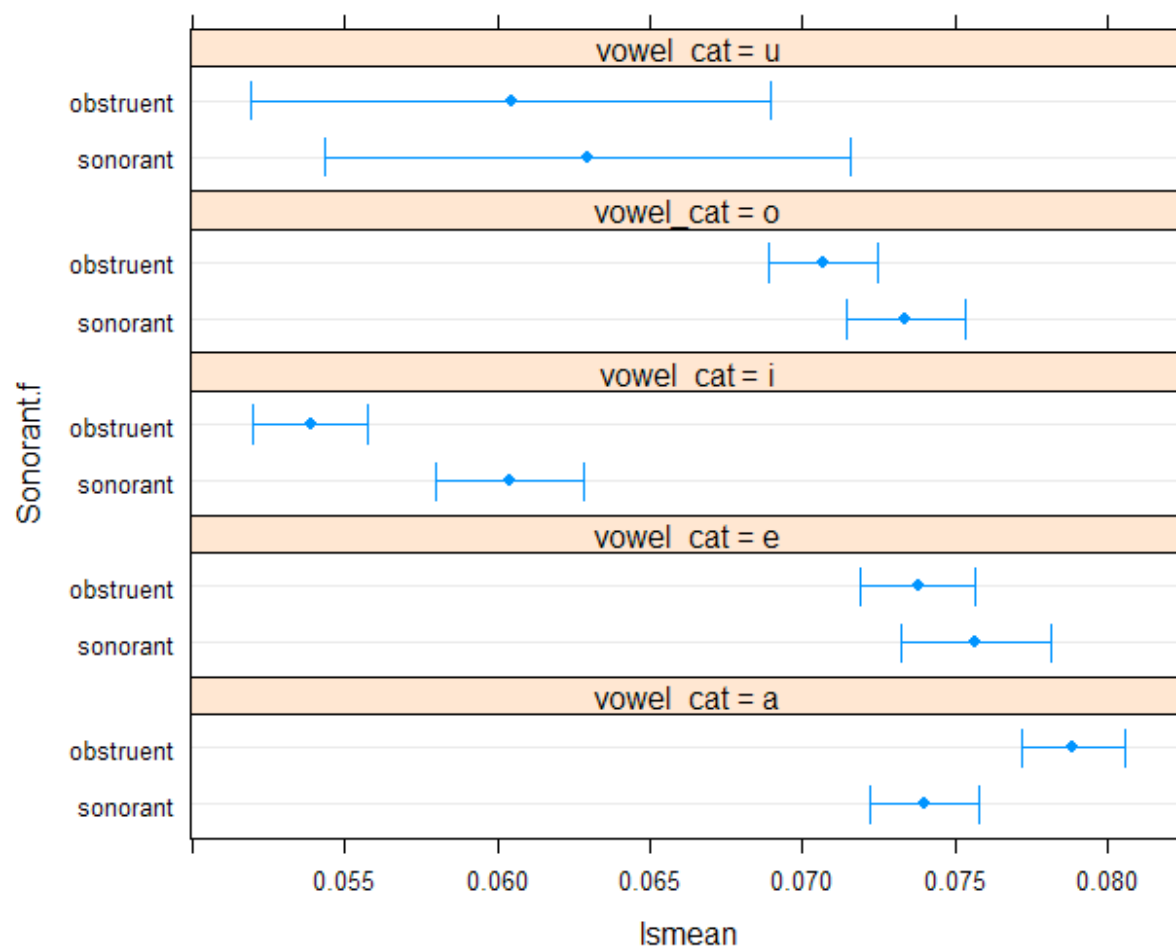
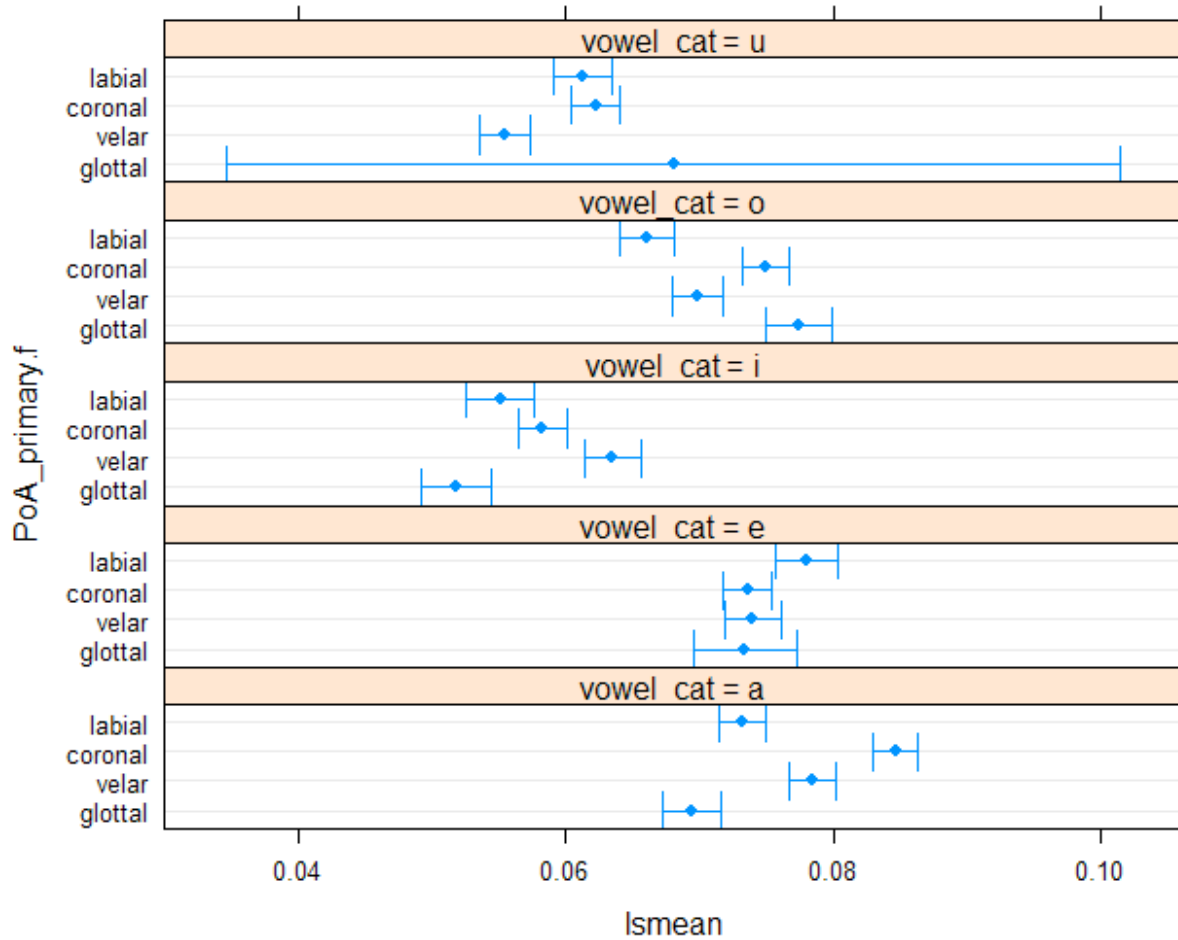


Figure 10: lsmeans for PLACE of articulation by vowel. Error bars indicate 95% confidence intervals.



Given the interactions between CVENTROPY and VOWEL, as well as the presence of other significant interactions between VOWEL and other factors (PLACE, SONORANCY), we sought to probe the effects of CVENTROPY further by fitting the full CVENTROPY model to each vowel separately. This analysis also afforded us the opportunity to add another factor, PALATAL, to the

vowel-specific models of /a/, /e/, /u/, and /o/.<sup>6</sup> PALATAL picks out preceding consonants with either a primary or secondary palatal articulation: /y/, /hy/, /sy/, /zy/. The factor was dummy coded with non-palatal segments as the reference category. Table 5 summarizes the vowel-specific models.

**Table 5: The CVEntropy model for each separate vowel.**

|                     | /a/          |             | /i/     |         | /u/          |             | /e/     |         | /o/           |              |
|---------------------|--------------|-------------|---------|---------|--------------|-------------|---------|---------|---------------|--------------|
| <b>Fixed Factor</b> | $\beta$      | t value     | $\beta$ | t value | $\beta$      | t value     | $\beta$ | t value | $\beta$       | t value      |
| <b>(Intercept)</b>  | 0.058        | 27.24       | 0.053   | 4.45    | 0.039        | 1.94        | 0.077   | 9.15    | 0.096         | 14.22        |
| <b>VOICING</b>      | -0.009       | -17.41      | -0.012  | -10.50  | -0.011       | -12.51      | -0.009  | -8.11   | -0.014        | -15.23       |
| <b>VELAR</b>        | 0.006        | 7.01        | 0.006   | 4.22    | -0.015       | -0.91       | -0.002  | -0.64   | -0.005        | -3.47        |
| <b>CORONAL</b>      | 0.015        | 16.99       | 0.004   | 2.73    | -0.013       | -0.77       | 0.000   | -0.20   | -0.001        | -0.73        |
| <b>LABIAL</b>       | 0.010        | 9.16        | 0.001   | 0.74    | -0.012       | -0.68       | -0.003  | -1.12   | -0.010        | -6.67        |
| <b>CVENTROPY</b>    | <b>0.004</b> | <b>5.67</b> | 0.003   | 0.50    | <b>0.015</b> | <b>2.75</b> | -0.003  | -0.77   | <b>-0.008</b> | <b>-2.63</b> |
| <b>PALATAL</b>      | -0.006       | -6.11       | NA      | NA      | 0.013        | 3.33        | -0.013  | -2.84   | -0.012        | -5.56        |
| <b>SONORANCY</b>    | 0.003        | 4.80        | -0.005  | -4.23   | -0.003       | -2.68       | 0.000   | -0.09   | -0.006        | -6.67        |
| <b>SYLL_STRUC</b>   | 0.014        | 26.99       | 0.011   | 14.00   | 0.013        | 16.25       | 0.017   | 21.64   | 0.016         | 20.05        |

Table 5:  $\beta$  estimates and t values for fixed factors in mixed models fit separately to each vowel.

Across vowels, the effect of consonant voicing was always in the same direction. The negative  $\beta$  estimate indicates that vowels are shorter when following voiceless consonants than when following voiced consonants. The size of the effect ranges across vowels from 9 ms (for /a/, /e/) to 14 ms (for /o/). PLACE of articulation also showed reliable effects. The direction of the PLACE of articulation effects vary across vowels. The  $\beta$  estimates for VELAR, CORONAL, LABIAL are mostly positive for /a/, /i/ and /u/ and negative for /e/ and /o/. This is due in part to the fact that

<sup>6</sup> We excluded PALATAL from the larger models (Tables 2 and 3) because the interaction with VOWEL was rank deficient (owing to the absence of /i/ following palatal consonants), and without the interaction term, the variance explained by PALATAL did not justify inclusion.

vowel duration following the glottals (the reference category for place) also varies substantially by vowel. As can be seen in Figure 3, /a/, /i/, /u/ are relatively short when following /h/ while /e/ and /o/ are relatively long. In particular, /a/ is shorter following /h/ than in any other consonantal context and shorter than the average duration of /e/ and /o/ after /h/. This vowel-specific patterning may be due to aerodynamic factors. Retraction of the tongue body for /a/ may narrow the pharyngeal cavity facilitating sustained turbulence for /h/, effectively delaying the onset of voicing. More generally, the vowels with narrower constrictions may have this effect to different degrees at different constriction locations: /a/, pharyngeal; /i/, hard palate; /u/ soft palate/uvula. The variation across vowel quality in baseline vowel duration for PLACE makes it more insightful to interpret the relative differences between the VELAR, CORONAL, and LABIAL levels than the  $\beta$  estimates in isolation. For /a/, vowel duration was longest after coronals, then labials and then velar and glottals: coronal > labial > velar (> glottal). Each of the other vowels showed a different pattern of PLACE effects. /o/ was shortest after labials followed by velars (/o/: (glottal), coronal > velar > labial). /i/ was longest after velars, followed by coronals (/i/: velar > coronal, (glottal), labial). The effects of place on /e/ and /u/ were not particularly robust. The place effects on /e/ were small and the effects on /u/ were large but unstable, a pattern that came through in the bigger model as well.

PALATAL had a large (12-13 ms) shortening effect on /o/ and /e/, a smaller (6 ms) shortening effect on /a/ and a lengthening effect (13 ms) on /u/. SONORANCY was significant for /a/, /i/, /u/ and /o/, but, as we also saw in the big model, the direction of the effect on /a/ (positive) is different from the other vowels. Finally, the effect of CVENTROPY also varied across vowels in both size and direction.

There were significant positive effects of CVENTROPY on /u/ (15 ms) and /a/ (4 ms) and a negative effect of ENTROPY on /o/. The front vowels, /e/ and /i/, were not similarly affected. Lastly,

we again evaluated the statistical significance of CVENTROPY on the individual vowels by model comparison. The baseline model differed from the model summarized in Table 5 only in that it lacked CVEntropy as a fixed factor. We compared by maximum likelihood tests the baseline model to the CVEntropy model for each vowel. The results are summarized in Table 6. The inclusion of the CVEntropy factor into the model led to significant improvement for three out of the five vowels: /a/, /i/, /o/ but not the front vowels /e/, /i/.

Table 6: Comparisons between the baseline model and the CVEntropy model using maximum likelihood tests for each vowel.

|     | <b>AIC<br/>Baseline<br/>model</b> | <b>AIC<br/>CVEntropy<br/>model</b> | <b>Chisq</b> | <b>Pr(&gt;Chisq)</b> |
|-----|-----------------------------------|------------------------------------|--------------|----------------------|
| /a/ | -343813                           | -343840                            | 29.76        | 4.902e-08 ***        |
| /i/ | -199964                           | -199962                            | 3.056        | 0.62                 |
| /u/ | -183962                           | -183967                            | 7.56         | 0.006**              |
| /e/ | -163181                           | -163061                            | 0.59         | 0.44                 |
| /o/ | -269858                           | -269863                            | 6.91         | 0.00858 **           |

#### 4. Discussion

To summarize the results, we found that CVENTROPY has a significant effect on vowel duration for three out of the five vowels. Thus, the improvement of the CVENTROPY model over the baseline model is largely attributable to how CVENTROPY improves predictions for vowels /a/, /u/, and /o/. As the contextual uncertainty of the vowel increases, /a/ and /u/ show increased duration while /o/ decreases in duration.

On the front vowels, CVENTROPY had no effect on duration. This lack of effect may be because front vowels have other ways to signal their presence besides lengthening. The front

vowels, /i/ in particular, have strong coarticulatory influences on preceding consonants (Okada 1999: 118). Although beyond the scope of our current inquiry, it may be that the degree to which front vowels influence the articulation of preceding consonants is conditioned by entropy. We make the cursory observation that consonants with increasing entropy tend to be those that are more susceptible to coarticulation effects of /i/ (i.e., low coarticulatory resistance). Coarticulation may have a similar influence on increasing phonetic redundancy as lengthening the vowel. Palatalized consonants, where we see effects of CVENTROPY on vowel duration for /a/ and /u/, exhibit a high degree of coarticulatory resistance (e.g. Recasens & Espinosa 2009). It seems then, that vowel duration adjustment as a function of informativity plays a significant role in a non-arbitrary subset of the Japanese phonological system.

The negative effect of CVENTROPY on /o/ duration also requires comment. Amongst the five Japanese vowels, /o/ occurs as a long vowel far more frequently than the other vowels. In the CSJ corpus investigated here, /o:/ occurs in 1872 unique words compared with just 941 for /e:/, 599 for /u:/, 432 for /i:/, and 421 for /a:/. Given the likelihood of long /o:/, short /o/ may resist lengthening in response to informativity. As vowel uncertainty increases, it may become more important for short /o/ to maintain perceptual distinctiveness from /o:/ than from the other Japanese vowels. Reducing the duration of short vowels in high entropy environments would be one way to do this. The essence of this explanation is that when speakers are uncertain about vowel quality, they start caring about the length contrast as well, especially if the long competitor is frequent. This prediction can be tested in other languages that have a length contrast on vowels.

We would like to close this section with a few methodological remarks. Our measure of informativity is somewhat unique. Inspired in part by the proposal that informativity may permeate all levels of linguistic organization (Jaeger 2010), we explored a measure of informativity localized



within the phonology. The domain over which we computed the average predictability of a vowel was restricted to a phonologically relevant unit, the CV mora, in Japanese, as opposed to a unit of meaning, such as the morpheme or word. A second methodological point is that we looked at each segment in our analysis separately, which is also somewhat unusual in the antecedent informativity literature. Most of the studies reviewed in the introduction have related phonetic duration to the predictability of the higher level units, e.g., words, phrases, etc., in which phonological units are embedded. Attempts to pinpoint informativity effects in particular segments have been rare (although see Hall et al., 2016). Recall that in the current work, the overall effect of entropy was negative (Tables 3 and 4). Upon closer inspection, however, it turned out that the relationship between informativity and vowel duration is not as straightforward as it first appeared, because it varies across vowels for what may be principled phonetic reasons consistent with the broader informativity hypothesis. Overall, this study highlights the importance of boring down to individual data, which may reveal the interplay of various principles, including informativity, that govern phonetic behavior.

## **5. Conclusion**

To conclude, the current analysis of the CSJ reveals that various factors affect vowel duration in Japanese. In addition to these effects, consonant-conditioned entropy (CVEntropy) affects vowel duration as well, which supports the informativity hypothesis, but its positive effect surfaces in limited environments. We offered some explanations for why vowel lengthening does not occur in certain high entropy environments; vowel lengthening may be prevented when the vowel length contrast becomes important or when the preceding consonant is susceptible to coarticulation with the vowel, which can be used as a cue to the presence of that vowel.

The current study shows that, even at the level of CV-interaction, the effects of informativity may influence phonetic patterns. The current finding offers a new piece of insight regarding how informativity affects our speech behavior. Recall from the literature review in the introduction that most previous work focuses on higher level semantic/discourse effects. We have established, at least partly, that even at the level of phonology where meaning is irrelevant, informativity, as measured by CVEntropy, can have a non-trivial effect. However, this conclusion should also be taken with caution, because the effects of CVEntropy are not straightforwardly positive, but instead interact with other factors in a complicated way.

What this paper shows is necessarily limited. We have shown that informativity, measured as consonant-conditioned entropy, has some positive effects on some of the vowels. We should bear in mind, however, that our measure is just one way to quantify informativity. Nevertheless, we have demonstrated that informativity defined even at a local phonotactic level may influence phonetic patterns.

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